EXPERIMENT 03

Implementation and Analysis of Spatial Filtering Techniques using Low-Pass and High-Pass Filters

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files

uploaded = files.upload()
```

```
Choose files No file chosen    Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Screenshot 2026-08-14 193119.png to Screenshot 2026-08-14 193119.png
```

```
image = cv2.imread("Screenshot 2026-08-14 193119.png", cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Image not found!")
else:
    print("Image loaded successfully!")
    print("Image shape:", image.shape)
```

```
Image loaded successfully!
Image shape: (776, 873)
```

```
plt.figure(figsize=(6, 5))

plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.show()
```




```
average_filter = cv2.blur(image, (5, 5))

plt.figure(figsize=(6, 5))
plt.imshow(average_filter, cmap="gray")
plt.title("Average Filter")
plt.axis("off")
plt.show()
```




```
gaussian_filter = cv2.GaussianBlur(image, (5, 5), 0)

plt.figure(figsize=(6, 5))
plt.imshow(gaussian_filter, cmap="gray")
plt.title("Gaussian Filter")
plt.axis("off")
plt.show()
```



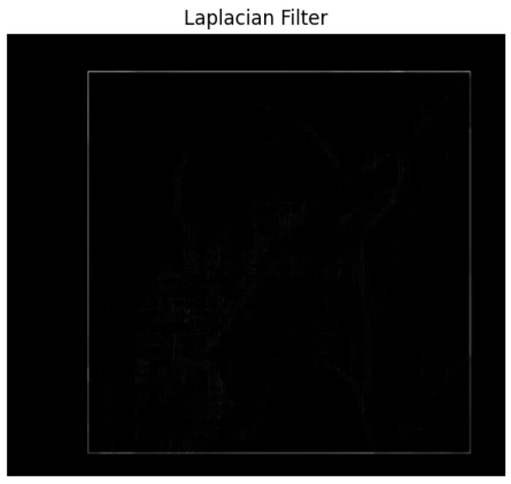

```
median_filter = cv2.medianBlur(image, 5)

plt.figure(figsize=(6, 5))
plt.imshow(median_filter, cmap="gray")
plt.title("Median Filter")
plt.axis("off")
plt.show()
```



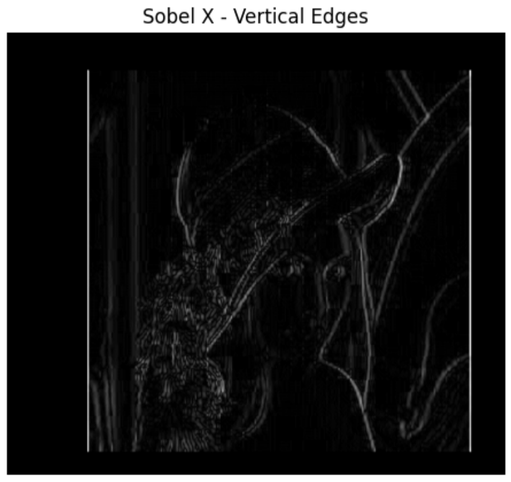

```
laplacian = cv2.Laplacian(image, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)
```

```
plt.figure(figsize=(6, 5))
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")
plt.show()
```




```
sobel_x = cv2.Sobel(image, cv2.CV_64F, 1, 0, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)

plt.figure(figsize=(6, 5))
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X - Vertical Edges")
plt.axis("off")
plt.show()
```



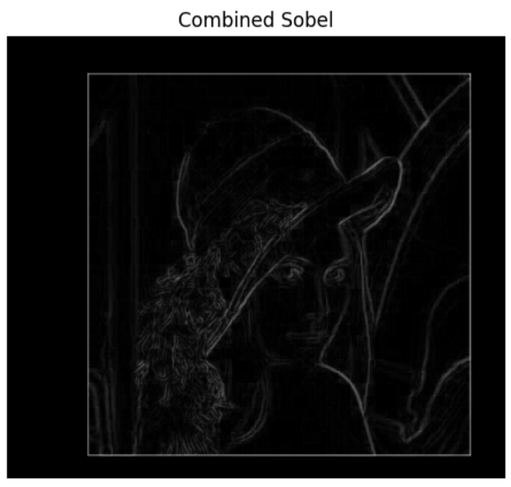

```
sobel_y = cv2.Sobel(image, cv2.CV_64F, 0, 1, ksize=3)
sobel_y = cv2.convertScaleAbs(sobel_y)

plt.figure(figsize=(6, 5))
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y - Horizontal Edges")
plt.axis("off")
plt.show()
```




```
sobel_combined = cv2.addWeighted(
    sobel_x, 0.5,
    sobel_y, 0.5,
    0
)

plt.figure(figsize=(6, 5))
plt.imshow(sobel_combined, cmap="gray")
plt.title("Combined Sobel")
plt.axis("off")
plt.show()
```




```
plt.figure(figsize=(14, 8))

plt.subplot(2, 4, 1)
plt.imshow(image, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(2, 4, 2)
plt.imshow(average_filter, cmap="gray")
plt.title("Average")
plt.axis("off")

plt.subplot(2, 4, 3)
plt.imshow(gaussian_filter, cmap="gray")
plt.title("Gaussian")
plt.axis("off")

plt.subplot(2, 4, 4)
plt.imshow(median_filter, cmap="gray")
plt.title("Median")
plt.axis("off")

plt.subplot(2, 4, 5)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian")
plt.axis("off")

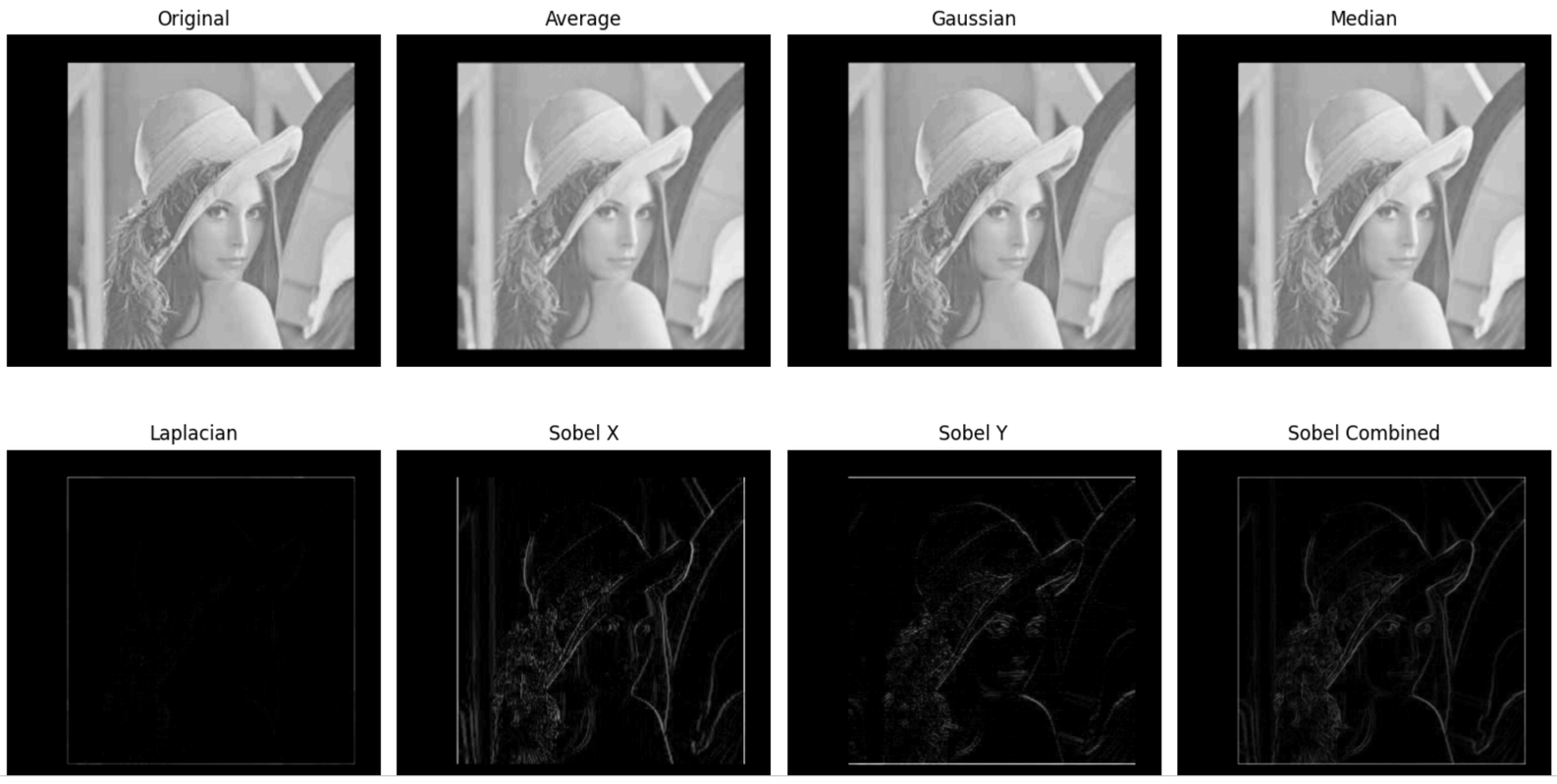
plt.subplot(2, 4, 6)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X")
```

```
plt.axis("off")

plt.subplot(2, 4, 7)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y")
plt.axis("off")

plt.subplot(2, 4, 8)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Sobel Combined")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Questions

Q1. What is spatial filtering? How is it used in digital image processing?

Answer: Spatial filtering is an image processing technique in which the value of a pixel is modified using its neighboring pixels. A small matrix called a kernel or mask is moved over the image.

Spatial filtering is used for:

Noise removal Image smoothing Sharpening Edge detection Image enhancement

Example: Gaussian filter is used for smoothing, while Sobel filter is used for edge detection.

Q2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples. Answer:

Low-Pass Filter High-Pass Filter Removes high-frequency components Enhances high-frequency components Used for smoothing Used for edge detection Reduces noise Highlights edges Produces a smoother image Enhances fine details Examples: Average, Gaussian, Median Examples: Sobel, Laplacian

Low-Pass ⟶ Smoothing and Noise Reduction

High-Pass ⟶ Edge Detection and Sharpening

Q3. Compare Average Filter, Gaussian Filter, and Median Filter based on noise reduction and edge preservation. Average Filter Answer: It calculates the average value of neighboring pixels. It reduces noise but can blur edges.

Gaussian Filter

It gives more importance to pixels near the center and provides smooth noise reduction.

Median Filter

It replaces a pixel with the median value of its neighborhood. It is especially effective for salt-and-pepper noise and preserves edges well.

Q4. Why is the Median Filter particularly effective for removing salt-and-pepper noise? Answer: Salt-and-pepper noise consists of random black and white pixels.

The Median filter replaces a pixel with the median value of its neighboring pixels. Extreme black and white values have less effect on the median.

Therefore, the Median filter removes salt-and-pepper noise while preserving important edges.

Q5. Explain the role of convolution in spatial filtering. Answer: Convolution is a mathematical operation used in spatial filtering. A small matrix called a kernel is moved over the image.

The kernel values are multiplied with the corresponding image pixels and then added to produce the new pixel value.

Q6. What is the purpose of Sobel and Laplacian operators in edge detection?

Answer: Sobel and Laplacian operators are High-Pass filters used for detecting edges.

Sobel Operator Sobel X detects vertical edges. Sobel Y detects horizontal edges. Laplacian Operator

The Laplacian operator uses the second derivative to detect rapid changes in intensity and edges in multiple directions.

Therefore:

Sobel ⟶ Directional edge detection

Laplacian ⟶ Multi-directional edge detection

Q7. Why are Gaussian filters considered smoothing filters in image processing?

Answer: Gaussian filters are smoothing filters because they reduce sudden changes in pixel intensity.

They assign higher weights to pixels near the center and lower weights to pixels farther away.

Gaussian filtering is used to:

Reduce noise Remove small details Smooth images Prepare images for edge detection

Q8. Benefits and limitations of frequency-domain filtering

Answer: Benefits:

Removes periodic noise effectively. Separates low and high frequencies. Useful for image enhancement.

Limitations:

More complex than spatial filtering. Requires Fourier Transform. Needs more computation.

Q9. Two real-world applications of spatial filtering

Answer:

Medical Imaging – Noise removal and image enhancement in X-rays, MRI, and CT scans. Computer Vision – Edge detection in face detection, CCTV, and autonomous vehicles.

Q10. Spatial-domain vs Frequency-domain filtering. Answer Spatial Domain Frequency Domain Works directly on pixels Works on frequency components Uses kernels/masks Uses Fourier Transform Simple and easy More complex Used for smoothing and edge detection Used for noise removal and enhancement

In short: Spatial ⟶ Pixels + Kernel Frequency ⟶ Fourier Transform